Cargo Robot: Parts Still Missing

Shopping list, 12 November 2026. The 3D-printed parts and the laser-cut plates (all **6 mm acrylic**) are done, and so is everything in the "already have" section, so everything below is **to buy**. Tick each box when you have it. Names and sizes match the build guide (section 0.2) and the drawings. **[changed]** marks what changed with the acrylic plates.

1. Buy these first: drive, battery and servo power

	#	Item	Qty	What to check / what it's for
☐	1	**JGB37-520 12 V 110 RPM gear motor SET**: motor with Hall encoder (6 wires) + steel L-bracket + hex coupling + **65 mm wheel**	2	One set per side, both the same RPM. Shaft is 6 mm with a flat (sellers often write "5.5 mm D-shaft"); the coupling must fit it. **2 wheels is all you need**: they drive and steer, and the 2 printed ball casters (front and rear) carry the rest.
☐	2	**Battery: 3S LiPo 11.1 V 5000 mAh**, 20 C or more, **XT60** plug	1 (2 is better)	"3S" = 3 cells. Check it has the small white balance plug. A second battery saves the demo day.
☐	3	**Buck converter XL4016 8 A**, with voltage pot and heat sink	1	**Needed**: it feeds the 2 servos at 7.4 V. The battery's 11-12.6 V would burn them (8.4 V max), and on your 5 A converter the servos would pull the voltage down and reset the ESP32 while the bed tips. Keep the 5 A one for the ESP32 and sensors.

2. Battery care

	#	Item	Qty	What to check / what it's for
☐	4	LiPo balance charger (IMAX B6 / B6AC class)	1	The plain B6 also needs a 12 V, 5 A or bigger power adapter; the B6AC has one built in.
☐	5	LiPo fire-safe bag	1	Charge and store the battery inside it.
☐	6	LiPo voltage alarm (1-8 S buzzer)	1	Plugs into the balance lead. Set it to 3.5 V per cell.

3. Panel: safety and controls (they mount through the 6 mm deck)

	#	Item	Qty	What to check / what it's for
☐	7	Emergency-stop button, 22 mm panel-mount, latching, with an **NC** contact block (10 A) **[changed]**	1	NC = normally closed: power flows until you press it. Check it clamps a **6 mm** panel.
☐	8	Toggle switch (master), 12 mm hole, rated **12 V DC, 15 A or more** **[changed]**	1	The DC rating matters, not the AC one. Thread at least **9 mm** long (6 mm deck + nut).
☐	9	Push button, 12 mm panel-mount, momentary, normally open **[changed]**	1	The start button. Thread at least **9 mm** long.
☐	10	5 mm LEDs, green + red, with 5 mm panel clips **[changed]**	1 each	Check the clips fit a 6 mm panel. If they don't, skip them and hot-glue the LEDs into the 6 mm holes from below.
☐	11	Active buzzer module, 3 pins, 3.3-5 V	1	"Active" = it beeps by itself when powered.

4. Power wiring and connectors

	#	Item	Qty	What to check / what it's for
☐	12	Inline blade-fuse holder (14 AWG) + 10 A blade fuses	1 + 5	
☐	13	XT60 connector pairs (male + female)	4	
☐	14	WAGO 221-415 lever connector (5-way)	8	Power buses without soldering.
☐	15	WAGO 221-413 lever connector (3-way)	6	
☐	16	Silicone wire 16 AWG, red + black	3 m each	Battery, fuse, switch, E-stop, motor drivers, servo supply.
☐	17	Silicone wire 20 AWG, red + black	2 m each	Buck converters, motors.

5. Signal wiring and small parts

	#	Item	Qty	What to check / what it's for
☐	18	Jumper wires, female-female and male-female, 20 cm	40 each	Signals.
☐	19	Servo extension leads, 30 cm	2	Servo to the I/O board.
☐	20	USB **data** cable for the ESP32 (micro-USB or USB-C, to match the board), 1 m or longer	1	Some cheap cables only charge: try it on the laptop first.
☐	21	Resistors 1/4 W: 1 kΩ × 3, 2 kΩ × 3, 10 kΩ × 5, 2.2 kΩ × 1, 330 Ω × 2	1 set	Voltage dividers, pull-ups, LEDs.
☐	22	Capacitor 100 nF (ceramic)	1	
☐	23	Perfboard 5 × 7 cm	2	The I/O board (1 spare).
☐	24	Ferrule kit + ferrule crimper	1	A ferrule on every wire that goes into a screw terminal.
☐	25	Heat-shrink assortment, zip ties, spiral wrap, 20 mm velcro strap (1 m), double-sided foam tape	1 each	
☐	26	VHB tape (strong double-sided foam), 19 mm	1 roll	Holds the small boards without drilling.

6. Rods, bolts, screws, nuts and washers (sizes checked for the 6 mm acrylic plates)

	#	Item	Qty	What to check / what it's for
☐	27	M6 threaded rod, 1 m, zinc-plated steel	1	Cut into 6 pieces of 130 mm.
☐	28	M6 nuts + M6 washers	26 each	24 needed (4 per rod).
☐	29	M8 × 50 bolt + 2 M8 nuts + 2 M8 washers	2 sets	One set per ball caster.
☐	30	M4 × 10	2	Bed rests.
☐	31	M4 × 12 **[changed]**	6	Servo mounts. **No washer** under the head here: that leaves about 4 turns of thread in the nut, plenty. **Not longer**: the tip would poke out of the foot into the servo.
☐	32	M4 × 16	4	Servo ear clamps.
☐	33	M4 × 20	12	Horn adapters (4) + risers from below (8).
☐	34	M4 × 20 **countersunk (flat head)**	8	Down through the cargo plate into the risers: flush, so the package slides over them.
☐	35	M4 nuts + M4 washers	32 each	
☐	36	M3 × 8	8	Disc horns onto the horn adapters. **Not longer**: longer screws hit the adapter's flange.
☐	37	M3 × 8 **countersunk**	6	Caster rings onto the cups.
☐	38	M3 × 12 + M3 nuts	11 each	Motor brackets (8) + IR modules onto the sensor bar (3). On the motor brackets, check the tip doesn't touch the motor.
☐	39	M3 × **16** + M3 nuts **[changed]**	4 each	Line sensor bar (2) + ultrasonic bracket (2): the printed part + 6 mm plate + nut. M3 × 12 is too short (only about 1 thread in the nut).
☐	40	M3 washers **[changed]**	30	Under the M3 heads and nuts on the bottom plate. It's acrylic, not plywood: washers spread the load so it doesn't crack.

7. Casters, track and finishing

	#	Item	Qty	What to check / what it's for
☐	41	Steel bearing ball, 25.4 mm (1 inch)	3	2 ball casters + 1 spare. A 25 mm glass marble also works.
☐	42	Black **matte** tape, **19 mm** wide	1 roll	The line. Not 50 mm: it would cover all 3 sensors and look like a station mark.
☐	43	Masking tape	1 roll	Laying out the track.
☐	44	Self-adhesive rubber pads, 18 mm, 2 mm thick	2	On the 2 bed rests.
☐	45	Blue thread-locker (medium strength)	1	Coupling set screws, horn screws.
☐	46	Grease (Vaseline is fine)	a little	Caster cups.

8. Because the plates are 6 mm acrylic

- All four plates are 6 mm acrylic. The first drawings had 6 mm *plywood* for the bottom and tilting plates and 5 mm acrylic for the deck and cargo plate, so only the deck and the cargo plate got 1 mm thicker. The drawings, the build guide and this list now all match. Items marked **[changed]** are the ones this affected.
- Almost nothing else changes: the rods stay 130 mm and the other screws are still long enough. The top of the cargo plate is about **220 mm** above the floor (was 218).
- Acrylic cracks more easily than plywood and bends about twice as much. The loads on this robot are small, so it's fine, but put washers under heads and nuts on the bottom plate, tighten only until snug, and check the rod, caster and motor-bracket holes for cracks after the first test drives.
- The only holes you drill yourself are the motor-bracket holes, now in acrylic. Use a sharp bit, drill slowly, with a piece of wood clamped underneath. Countersink the cargo plate's 8 holes slowly too.
- **Two fixes to the old list** (not caused by the acrylic): the M3 × 12 screws for the line sensor bar and the ultrasonic bracket were always too short (now M3 × 16), and the servo-horn screws are M3 × 8, not M3 × 10.

9. You already have

- ☑ All 3D-printed parts and all laser-cut plates (6 mm acrylic)
- ☑ ESP32 DevKitC 38-pin + 38-pin screw-terminal breakout board
- ☑ 2 × BTS7960 (IBT-2) motor drivers
- ☑ XL4015 5 A buck converter (for the ESP32 and sensors, set to 5.1 V)
- ☑ 2 × DS5160 servos + 2 round metal disc horns
- ☑ Sensors: the 3-pin IR line modules (3 + 1 spare) and the HC-SR04 ultrasonic sensor
- ☑ Tools (soldering iron, multimeter, drill and bits, spanners, ...)

10. Questions for you: what else might be missing?

Q1 **Your disc horns:** are they 38 mm across or less, with 4 screw holes on a circle 19-28 mm across? The printed horn adapter's slots only fit that range.

Q2 **Spares:** do you have a second ESP32 and a second 5 A buck converter? The build guide suggests one spare of each (a wrong wire can kill either one, and the shop may be days away).

Q3 **The floor for the track:** is it white or light-coloured? The IR sensors need black tape on a light floor. If the floor is dark, add white tape, white boards or a roll of white vinyl.

Q4 **Panel parts you may already own:** if you already have an E-stop, switch or button, will its thread reach through the 6 mm deck? If not, it needs replacing or a thinner spacer plate.